HW5 (Chapter 9) for MSE 201:002 Spring 2023

Textbook:
W.D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8thEdition

Chapter 9

9.13 For an alloy of composition 74 wt% Zn–26 wt% Cu, cite the phases present and their compositions at the following temperatures: 850°C, 750°C, 680°C, 600°C, and 500°C.

9.26 It is desirable to produce a copper–nickel alloy that has a minimum noncold-worked tensile strength of 350 MPa (50,750 psi) and a ductility of at least 48%EL. Is such an alloy possible? If so, what must be its composition? If this is not possible, then explain why.

9.32 For a copper–silver alloy of composition 25 wt% Ag–75 wt% Cu and at 775°C (1425°F), do the following:

(a) Determine the mass fractions of α and β phases.

(b) Determine the mass fractions of primary α and eutectic microconstituents.

(c) Determine the mass fraction of eutectic α.

9.57 Compute the maximum mass fraction of proeutectoid cementite possible for a hypereutectoid iron–carbon alloy.

9.63 For an iron–carbon alloy of composition 5 wt% C–95 wt% Fe, make schematic sketches of the microstructure that would be observed for conditions of very slow cooling at the following temperatures: 1175°C (2150°F), 1145°C (2095°F), and 700°C (1290°F). Label the phases and indicate their compositions (approximate).